

Jinsuk Mo



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Korea University (Sungbuk-gu, Seoul, Republic of Korea)
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EDUCATION

Department of Materials Science and Engineering, Korea University *Mar. 2023 – Present*
(4-semester hiatus due to mandatory military service)
Bachelor Student
Expected graduation: Feb. 2029
Culminative GPA: 4.42/4.5, Major GPA: 4.28/4.5

RESEARCH INTERESTS

- Material Characterization
- Phase Transformations
- Surface and Interface Characteristics
- MXene composites: catalytic performances, defects, and morphologies

RESEARCH EXPERIENCE

Undergraduate Research Assistant (PI: Prof. Yun-Mo Sung), Korea University *Mar. 2024 – Nov. 2024*

- Single-atom Electrocatalyst
 - Design single-atom electrocatalyst with N₄ moieties for efficient 4-electron pathway Oxygen Reduction Reaction (ORR)

CONFERENCE PARTICIPATION (*: Contributed equally, †: Corresponding author)

- Suh, J.*, Lee, G.*, **Mo, J.***, Kim, G.*, Chun, Y.*, Kim, H.*, Kim, S.*, and Lee, M†, “An innovative strategy for the synthesis of single-atom catalysts on MXene surfaces”, 2024 *Fall Meeting of the Korean Ceramic Society* (17/10/2024)
 - Abstract book p. 307., PG3-8, Energy Environmental Ceramics

HONORS AND AWARDS

- **Honorable Mention** *Mar. 2025 – Sep. 2025*
The 6th Corrosion Case Poster Contest 2025
The Corrosion Science Society of Korea
 - **Research Topic:** Formation of Goethite layer for preventing rebar corrosion in marine atmosphere
- **Co-recipient of the 1st Place Award, Excellence Award** *Aug. 2024 – Oct. 2024*
Smart Convergence Materials Science and Technology Competition
Gwangju Institute of Science and Technology (GIST)
 - **Research Topic:** Pd single-atom-coordinated g-C₃N₄ self-assembled partially oxidized Ti₃C₂ MXene type II-II heterojunction photocatalyst for reduction and conversion of greenhouse gas
 - Team Leader, Presenter
 - **Advisor:** Se-Ho Kim (Department of Materials Science and Engineering, Korea University)
- **3rd Place Award, Excellence Award** *Aug. 2024*
2nd Carbon Neutral Ideathon with Korea South-East Power Co. Ltd.
Korea South-East Power Co. Ltd. With participant universities in the Leaders in Industry-University Cooperation 3.0 (LINC 3.0): **Korea University, Kangwon National University, and Pohang Institute of Science and Technology (POSTECH)**
 - **Research Topic:** Facility for microalgae-based energy production utilizing unused land after the

- modernization of the power plant
- Team Leader, Presenter
- **Advisor:** Sanghun Sul (*Applied Innovation Experience Design (AIEX), Sungkyunkwan University*)
- **Dean's List** **Fall 2023**
- **Semester High Honors** **Fall 2024, Spring 2024, Fall 2023, Spring 2023**

Ongoing Project

- Campus Patent Universiade **Apr. 2026 -**
Korea Invention Promotion Association
 - **Research Topic:** Patent Landscape and Technology Roadmap of HfO₂-based Ferroelectric memories
 - Investigated a patent landscape and technology roadmap with a focus on 3D vertical architectures
 - Proposed technology strategies including interface reduction, oxygen-vacancy defect engineering, and MFS-compatible channel materials

SCHOLARSHIPS

- **SPECIAL SCHOLARSHIPS** **Fall 2024, Fall 2023**
Ministry of Student Affairs, Korea University
 - Recognized for past achievements and future potential
- **BEST HONORS SCHOLARSHIPS** **Spring 2024**
College of Engineering, Korea University
 - Recognized for academic excellence and outstanding performance
- **Daesung Mimac Scholarhips** **Spring 2023**
Digital Daesung
 - Scholarship for students with superior College Scholastic Ability Test (CSAT) score

MAJOR RELATED EXPERIENCES

Kinetic Investigations via Cyclic Voltammetry **Dec. 2024 – Jan. 2025**

- **Research Topic:** Quinone/Hydroquinone redox reactions on a carbon paper electrode in sulfuric acid electrolyte
 - Analyzed hydrophilic surface functionalization of carbon paper via cyclic voltammetry
- **Instructor:** Woong Kim (*Department of Materials Science and Engineering, Korea University*)

SK Hynix Campus Meet Up Day

Aug. 2024

SK Hynix

- Participant in a corporate training program of Processor Engineer role at SK Hynix
- <https://www.skcareersjournal.com/m/3328>

ABLEconomy for Sustainability MAKEathon

Jul. 2024

Brain Korea 21 Four R&E Center for Sustainable Living System and University of Cambridge Centre for Industrial Sustainability

- **Research Topic:** Recycling of glass into thermal insulation glass fiber
 - Explained the environmental significance of the supercooled state properties of glass in recycling
- Team Leader, Presenter
- **Advisor:** Jungmin Han (*Center for Human Ecology Research, Korea University*), and Kyuri Park (*Centre for Industrial Sustainability, University of Cambridge*)

Alchemist Future Innovation Idea Competition

Feb. 2024 – May. 2024

Ministry of Trade, Industry and Energy and Korea Planning & Evaluation Institute of Industrial Technology (KEIT)

- Government-funded project aimed at developing foundational technologies
- **Research Topic:** Reversible meta-surfaces and IR laser for optical communication
- Team Member, Presenter
- **Advisor:** Yongnam Song (*Department of Mechanical Engineering, Korea University*)

Samsung Shining Star

Jan. 2024

Samsung Electronics DS

- Participant in a corporate training program

EXTRACURRICULAR ACTIVITIES

Korea University AA CO-E Creator Club

Jul. 2024 – Present

- 1st Semester: Acquired basic knowledge of Python and VBA
Worked on a team project based on a random number generation algorithm
- Transistor Education and Hands-on Training** **Feb. 2025**
Korea Semiconductor Industry Association (KISA)
- Completed training course on semiconductor devices and processes, including transistor circuit design and electrical characteristic analysis (for 30 hours, 5 days)
- Understanding of the Power Generation** **Aug. 2024**
Korea Institute of Power Human Resource Development (KIPHRD)
- Completed training course on power plant, including gas turbine system, HRSG system, renewable energy, and energy transport (for 10 hours, online)
- Nanoscale Fabrication and Manufacturing, K-MOOC** **Aug. 2024**
Pohang Institute of Science and Technology (POSTECH) and K-MOOC
- Acquired advanced knowledge of semiconductor processing (Score: 86.67/100)
- Big Data Training and Practice** **Jul. 2024**
Innovation Center for Engineering Education, Korea University
- Completed training course on rudimentary data analysis using python (for 40 hours, 5 days)
- Introduction to Secondary Battery** **Jul. 2024**
Innovation Center for Engineering Education, Korea University
- Completed training course on electrochemistry for battery, including electrode materials and electrolyte materials (for 40 hours, 5 days)
- Reciprocal Teaching** **Mar. 2024 – Jun. 2024**
- Studied a course *Introduction to Ceramics* collaboratively

SKILLS AND TECHNIQUES

Experimental Techniques

- **Electrochemistry:** cyclic voltammetry (kinetic investigation and specific capacitance)
- **Finite Difference Method:** coursework for course *Kinetics for Materials Science*

Certificates

- TOEFL 110/120 (ETS, 2026)
- Goethe Zertifikat B1 (Goethe Institut, 2026)
- Computer Specialist in Spreadsheet and Database (Korea Chamber of Commerce and Industry, 2025)
- Advanced Data analysis Semi-Professional (K-Data, 2024)
- Driver's license (2022)

Languages

- Korean (Native), English (Full Academic fluency, CEFR C1), German (CEFR B1)

VOLUNTEER ACTIVITIES

- Donation for education** **Jan. 2025 – Mar. 2025**
NAVER and Korea Foundation for Science and Creativity (KOSAC)
- Tutored students with study questions online (for 5 hours)
- Korean language assistant for international students** **Jul. 2024 – Aug. 2024**
Korean Language Center, Korea University
- Tutored 2 international students Korean with activities and assisted with campus life (for 22 hours)
- Provided smartphone usage training for elders** **Jan. 2024 – Aug. 2024**
Lifeline Seoul
- Supported elders who are not familiar with smart devices (for 15 hours)

SELECTED EXPERIMENT BEFORE KOREA UNIVERSITY

A study of the microbial contamination of the faucet surface in the main building

Bugil Academy (Chungcheongnam-do 31115, Korea)

- Identification of *Staphylococcus Aureus* from a water faucet via MSA badge, PCR, Electrophoresis and UV Transilluminator
- **Advisor:** Jun-Young Kim (*Department of Microbiology, Dankook University, as of 2020*)